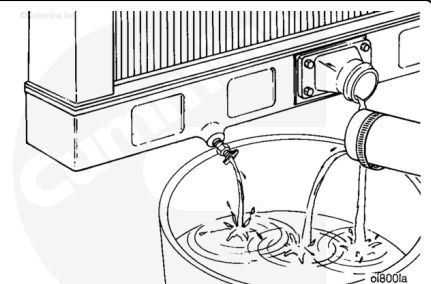


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Allow the engine to cool before draining to reduce the possibility of burns from hot liquid.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)

Remove

Industrial Applications

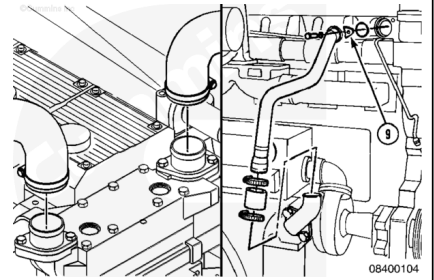
Conventional Aftercooling:

Remove both upper radiator hoses from the thermostat housing.

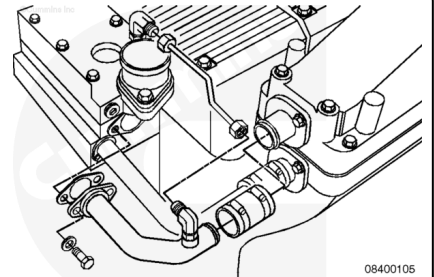
Remove the coolant bypass tube clip (9).

Loosen both hose clamps.

Remove the coolant bypass tube.



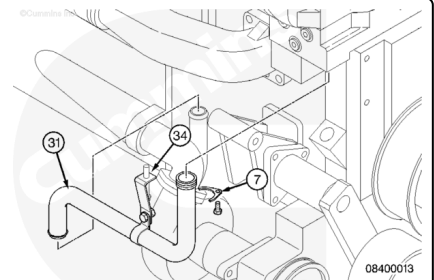
Remove the aftercooler coolant return tube, aftercooler coolant supply tube, gaskets, and hoses from the thermostat housing support.



Remove the aftercooler coolant supply tube clip (7). Loosen the hose clamps. Remove the support bracket (34).

Remove the tube (31).

Remove and discard the o-ring.

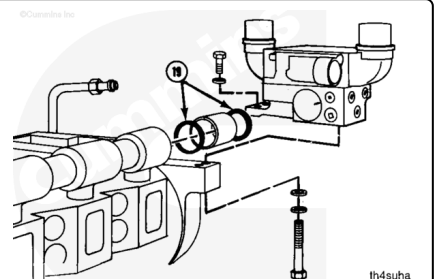


Disconnect the air compressor coolant outlet tube from the thermostat housing.

Remove the four mounting capscrews and the thermostat housing assembly.

Remove the coolant transfer tube.

Remove and discard the two o-rings (19).

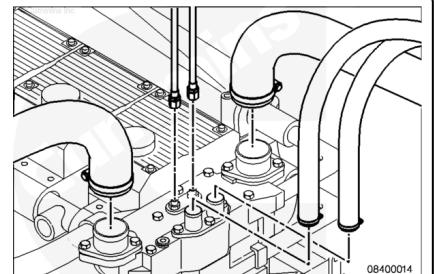


Low Temperature Aftercooling (LTA):

Remove both of the upper radiator hoses from the thermostat housing.

Remove both of the upper LTA radiator hoses from the thermostat housing.

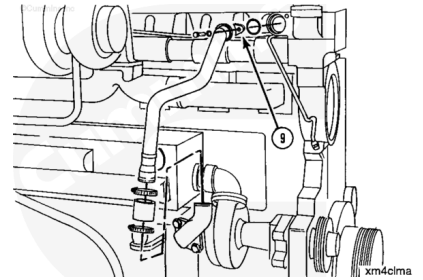
Remove both of the vent hoses from the thermostat housing.



Remove the coolant bypass tube clamp (9).

Loosen both hose clamps.

Remove the coolant bypass tube and hose.



Remove the water inlet and outlet tube clips (7).

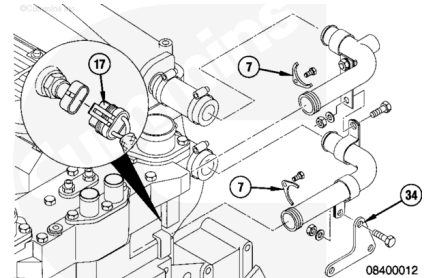
Loosen the hose clamps.

Remove the aftercooler coolant tube support bracket (34).

Remove the aftercooler coolant tubes.

Remove and discard the o-rings.

Disconnect the coolant temperature sensor wire (17).



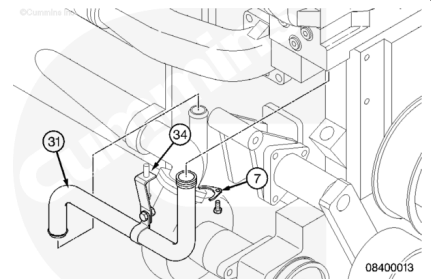
Remove the aftercooler coolant supply tube clip (7).

Loosen the hose clamps.

Remove the support bracket (34).

Remove the tube (31).

Remove and discard the o-ring.

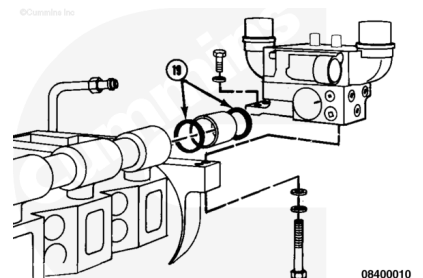


Remove the four mounting capscrews.

Remove the thermostat housing assembly.

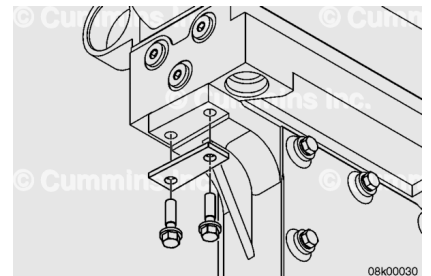
Remove the coolant transfer tube.

Remove and discard the two o-rings (19).

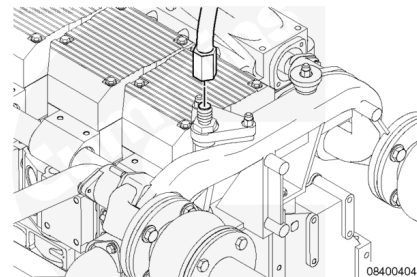


Note : Spacers are used on cast iron gear covers.

Remove the capscrews and spacer.

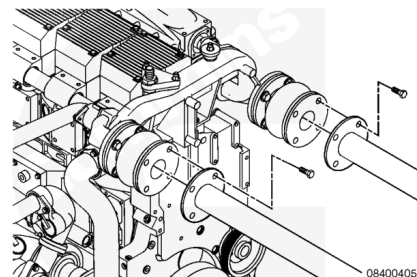


Disconnect the coolant makeup and vent hoses.



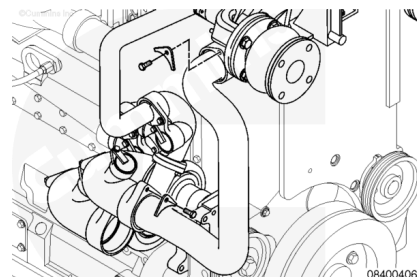
Disconnect the keel cooler supply and return pipes.

Remove the flexible connections, if necessary.

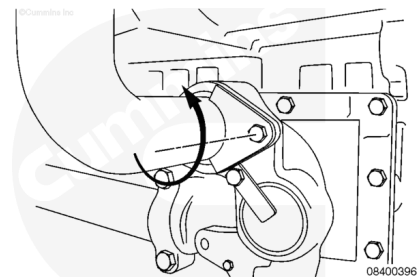


Remove the retaining clips from the water pump inlet pipe at the water pump and at the thermostat housing.

Remove the water pump inlet pipe.

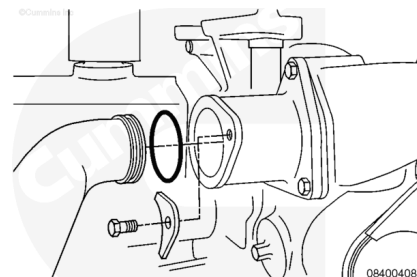


Loosen the retaining clip on the lower end of the LTA supply pipe.



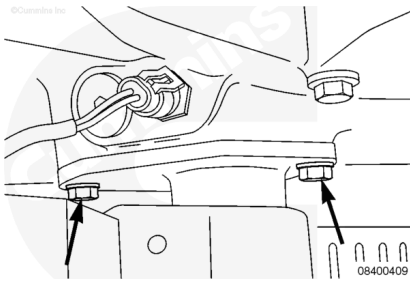
Remove the retaining clip at the thermostat housing and pull the LTA supply pipe out of the thermostat housing.

The pipe will rotate around the lower end.

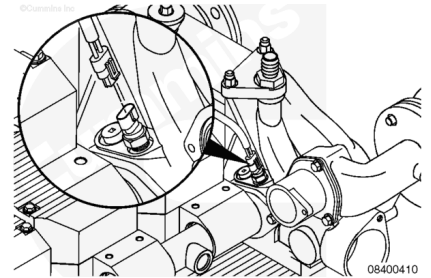


Remove the two capscrews holding the water rail flange to the thermostat housing.

Discard the o-ring.

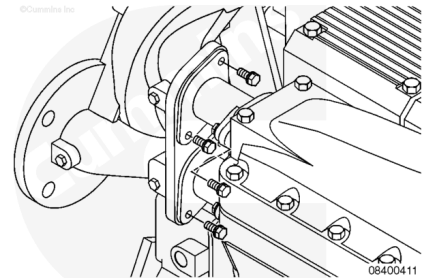


Disconnect the coolant temperature sensor.



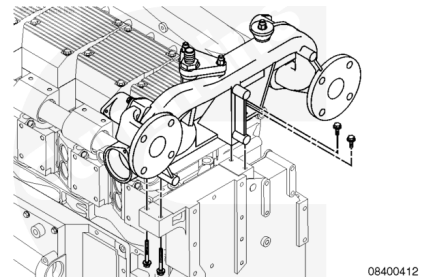
Remove the four capscrews from the aftercooler supply and return tubes.

Discard the o-rings.



Remove the four capscrews holding the thermostat housing to the gear cover.

Remove the thermostat housing.



Install

Industrial Applications

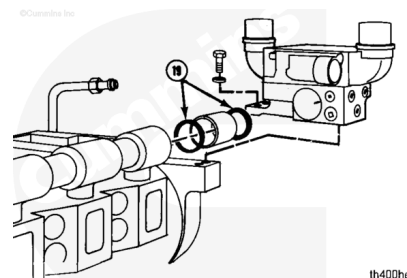
Conventional Aftercooling:

Install the o-rings (19) on the coolant transfer tube.

Use vegetable oil to lubricate the o-rings.

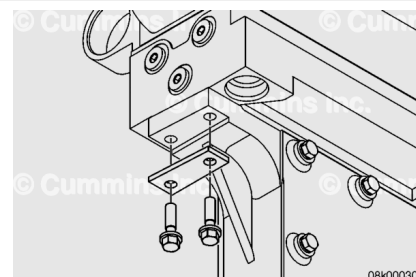
Install the tube in the bore of the rocker lever housing.

Align the bore in the thermostat housing support with the transfer tube. Push the support into position.



Spacers are used on cast iron gear covers.

Torque Value: 45 n•m [33 ft-lb]

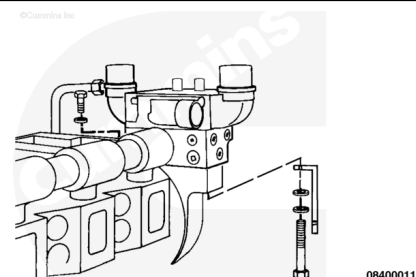


Note : The outside capscrew holds a clip for the aftercooler coolant inlet tube.

Install the four lockwashers and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Note : The raised lip on the retainer (7) must point toward the thermostat housing.

Use vegetable oil to lubricate the o-ring. Install a new o-ring on the aftercooler coolant supply tube (31).

Install the tube in the bore of the thermostat housing.

Install the retainer clip (7) and capscrew. Tighten the capscrew.

Torque Value:

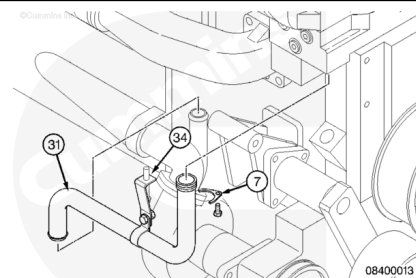
Capscrew 20 n•m [177 in-lb]

Tighten the hose clamps.

Torque Value:

Hose Clamp 6 n•m [53 in-lb]

Attach the support clip to the bracket (34).



Connect the aftercooler coolant return tube, the aftercooler coolant supply tube, gaskets, and hose(s) to the aftercooler.

Install the gasket and return tube capscrews.

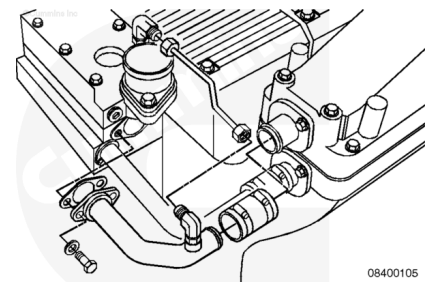
Tighten the capscrews.

Torque Value: 6 n•m [50 in-lb]

Install the vent hose.

Connect the coolant temperature sensor wire.

Note : If the engine is equipped with an air compressor, install the air compressor coolant return tube.



Use vegetable oil to lubricate the o-ring on the coolant bypass tube.

Install the coolant bypass tube.

Install the retainer (9) and capscrew.

Tighten the capscrew.

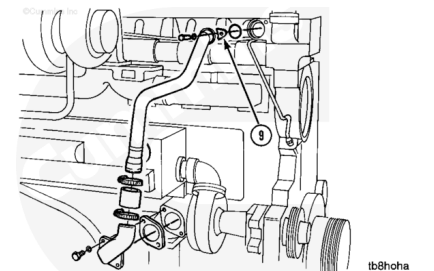
Torque Value:

Capscrew 45 n•m [33 ft-lb]

Tighten the hose clamps.

Torque Value:

Hose Clamp 6 n•m [50 in-lb]

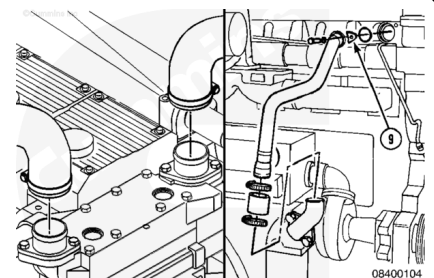


Install the two upper radiator hoses.

Tighten the hose clamps.

Torque Value: 6 n•m [50 in-lb]

Install the vent lines.



LTA:

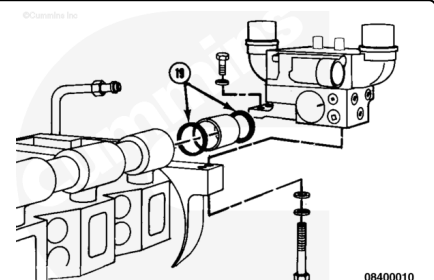
Install the o-rings (19) on the coolant transfer tube.

Use vegetable oil to lubricate the o-rings.

Install the tube in the bore of the rocker lever housing.

Align the bore in the thermostat housing support with the transfer tube.

Push the support into position.

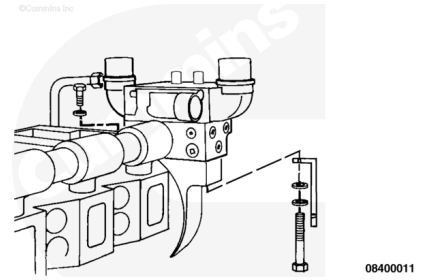


Note : The outside capscrew holds a clip for the aftercooler water inlet tube.

Install the four lock washers and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Use vegetable oil to lubricate the o-ring on the coolant bypass tube.

Install the coolant bypass tube.

Install the retainer (9) and capscrew.

Tighten the capscrew.

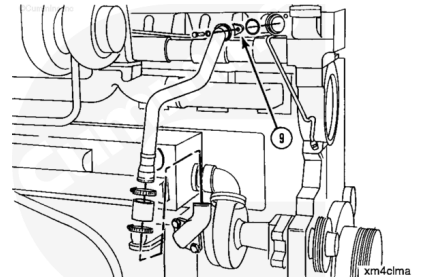
Torque Value:

Capscrew 45 n•m [33 ft-lb]

Tighten the hose clamps.

Torque Value:

Hose Clamp 6 n•m [50 in-lb]



Note : The raised lip on the retainer (7) must point toward the thermostat housing.

Use vegetable oil to lubricate the o-ring.

Install a new o-ring on the aftercooler coolant supply tube (31).

Install the tube in the bore of the thermostat housing.

Install the retainer clip (7) and capscrew.

Tighten the capscrew.

Torque Value:

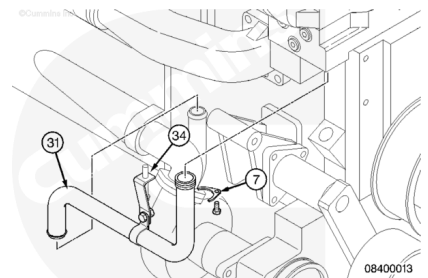
Capscrew 20 n•m [177 in-lb]

Tighten the hose clamps.

Torque Value:

Hose Clamp 6 n•m [50 in-lb]

Attach the support clip to the bracket (34).



Note : The raised lip on the retainer (7) must point toward the thermostat housing.

Use vegetable oil to lubricate the new o-rings.

Install the o-rings on the aftercooler coolant inlet and outlet tubes.

Install the tubes. Install the retainer clips (7) and capscrews.

Tighten the capscrews.

Torque Value:

Capscrew 20 n•m [177 in-lb]

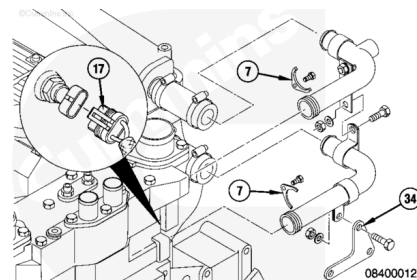
Tighten the hose clamps.

Torque Value:

Hose Clamp 6 n•m [50 in-lb]

Attach the support clips to the bracket (34).

Connect the coolant temperature sensor wire (17).



Install both of the radiator vent lines.

Install the upper low temperature aftercooling radiator hose.

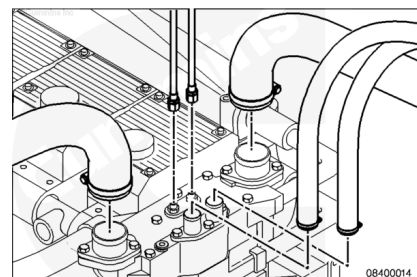
Tighten the clamps.

Torque Value: 6 n•m [50 in-lb]

Install both of the upper engine radiator hoses.

Tighten the clamps.

Torque Value: 6 n•m [50 in-lb]



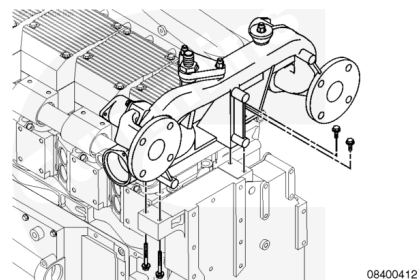
Marine Applications

Install the o-rings at the water rail flange and the aftercooler supply and return tubes.

Install the thermostat housing to the top of the gear cover. Use four capscrews.

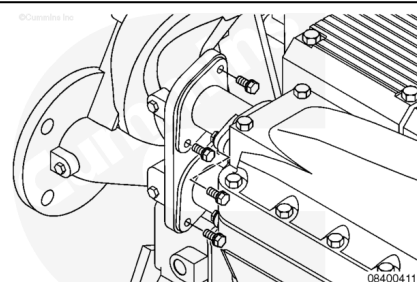
Make sure the o-rings are in place.

Torque Value: 40 n•m [30 ft-lb]



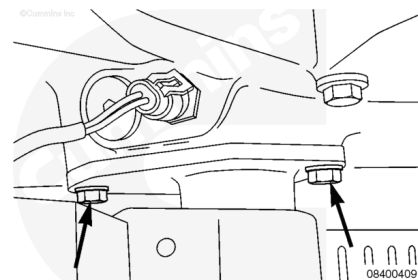
Install the four capscrews at the aftercooler supply and return tubes.

Torque Value: 40 n•m [30 ft-lb]

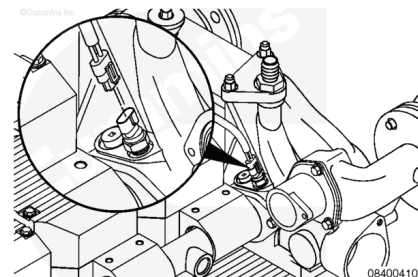


Install the two capscrews at the water rail flange to the thermostat housing.

Torque Value: 40 n•m [30 ft-lb]



Connect the coolant temperature sensor connector.

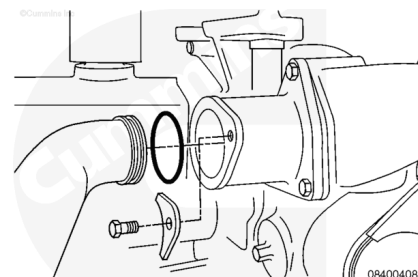


Install a new o-ring on the LTA supply pipe and insert into the thermostat housing.

Install the retaining clip.

Torque the upper and lower retaining clips.

Torque Value: 40 n•m [30 ft-lb]

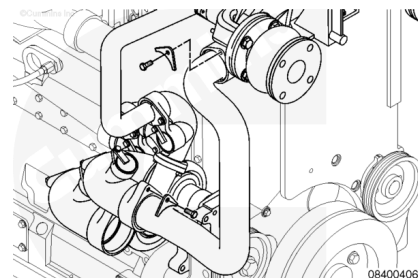


Install new o-rings on the water pump inlet pipe. Insert the inlet pipe into the water pump inlet and thermostat housing at the same time.

Install the retaining clips.

Torque the upper and lower retaining clips.

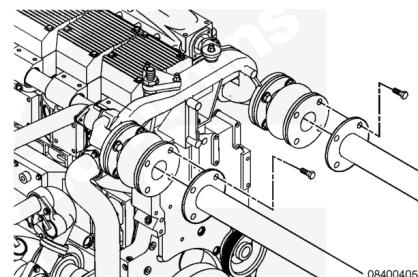
Torque Value: 40 n•m [30 ft-lb]



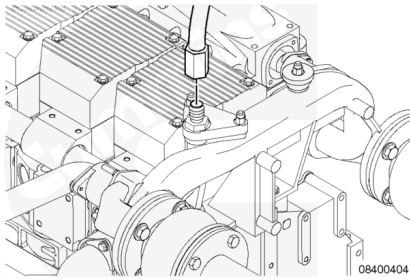
Install the flexible connections, if removed. Use four capscrews.

Torque Value: 180 n•m [135 ft-lb]

Connect the keel cooler supply and return pipes.



Connect the coolant makeup and vent hoses.



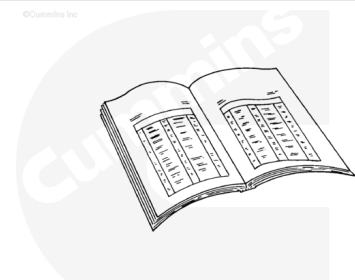
Finishing Steps

⚠ CAUTION ⚠

Use caution when filling the cooling system that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Start and operate the engine to 70°C [160°F] coolant temperature. Check for leaks.

©Cummins Inc.



ck800wa